



Sinhgad Academy of Engineering Department of Information Technology



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BE Project review No : 4

On

Manasarathi AI-Powered Mental Wellness Platform with Holistic Healing and Emotional Support

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Introduction

Introduction:

Mental health issues such as anxiety, depression, stress, and emotional imbalance are on the rise, especially in fast-paced, technology-driven lifestyles. Many people lack easy access to professional help, while others prefer self-paced, private wellness solutions. Our AI-Powered Mental Wellbeing App bridges this gap by combining modern psychological practices with ancient wellness techniques to guide users toward improved mental and emotional health.

AIM AND OBJECTIVE

Aim:

To create an AI-driven mobile application that assesses, guides, and supports users in improving their mental wellbeing using a blend of Western therapy and Eastern wellness approaches.

Objectives:

- To integrate holistic healing practices like meditation, mindfulness, and stress management.
- To design an empathetic chatbot capable of understanding and responding to human emotions.
- To monitor users' mood patterns and offer data-driven wellness recommendations.
- To ensure privacy, safety, and ethical use of AI in mental health assistance.

PROBLEM IDENTIFIED

- Limited access to affordable, quality mental health care.
- Stigma around seeking professional therapy.
- Overwhelming amount of unverified self-help content online.
- Lack of personalized, culturally sensitive guidance.
- Inability to track long-term progress in mental wellbeing.

LITERATURE SURVEY

SR. NO	REFERENCE NAME	SEED IDEA / WORK DESCRIPTION	PROBLEM FOUND
1.	Jorge A. Ruiz-Vanoye et al. (2025), <i>Artificial Intelligence and Human Well-Being: A Review of Applications and Effects on Life Satisfaction through Synthetic Happiness</i>	Reviews how AI influences life satisfaction through synthetic happiness, covering health, education, and social applications.	Lack of empirical evidence on long-term effects of synthetic happiness through AI.
2.	Anoushka Thakkar, Ankita Gupta, Avinash De Sousa (2024), <i>Artificial Intelligence in Positive Mental Health: A Narrative Review</i>	Explores AI's positive role in mental health such as diagnosis, emotional regulation, schizophrenia and neurodevelopmental disorders.	Existing AI models lack personalization and cultural sensitivity; ethical concerns raised.
3.	Zandra Bonnine V. Salcedo et al. (2024), <i>Artificial Intelligence and Mental Health Issues: A Narrative Review</i>	Narratively reviews AI's support for mental health treatments, including tools used in different countries.	AI-based solutions are underutilized due to lack of technical infrastructure and ethical standards.

LITERATURE SURVEY

SR. NO	REFERENCE NAME	SEED IDEA / WORK DESCRIPTION	PROBLEM FOUND
	Anish Marade, Kartik Suvarna, Prasad Chaudhari, Vinita Bhandiwad, <i>“Spiritual.AI: An AI-Driven Platform for Enhancing Mental, Emotional, and Spiritual Well-Being,”</i> IJRASSET, Vol. 13, Issue IV, Apr 2025	This paper introduces <i>Spiritual.AI</i> , a GPT-based AI platform providing personalized mental, emotional, and spiritual guidance. It integrates meditation, yoga, astrology, and productivity tools like the Pomodoro timer, combining ancient spiritual wisdom with modern AI for holistic well-being.	The system is conceptually strong but lacks large-scale implementation and real-world testing. Limited validation of AI’s effectiveness in handling deep psychological or spiritual issues.
	Kathleen Kara Fitzpatrick, Alison Darcy, Molly Vierhile, <i>“Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial,”</i> JMIR Mental Health, 2017.	This paper presents <i>Woebot</i> , an AI-powered conversational agent designed to deliver Cognitive Behavioral Therapy (CBT) techniques to young adults suffering from depression and anxiety. The chatbot interacts daily with users, offering emotional support and CBT-based mental health guidance.	The system focuses mainly on text-based CBT without deep personalization or spiritual integration. It lacks multimodal emotional recognition and adaptive therapy mechanisms

Objectives Finalized

- Simple, quick onboarding process.
- Multiple test options (personality, anxiety, stress, overthinking, trauma, emotional intelligence, etc.).
- Three-path solution: **Western**, **Eastern**, or **Hybrid** approach.
- AI-driven therapeutic conversations.
- Curated, quality content recommendations.
- Mood tracking and progress visualization.

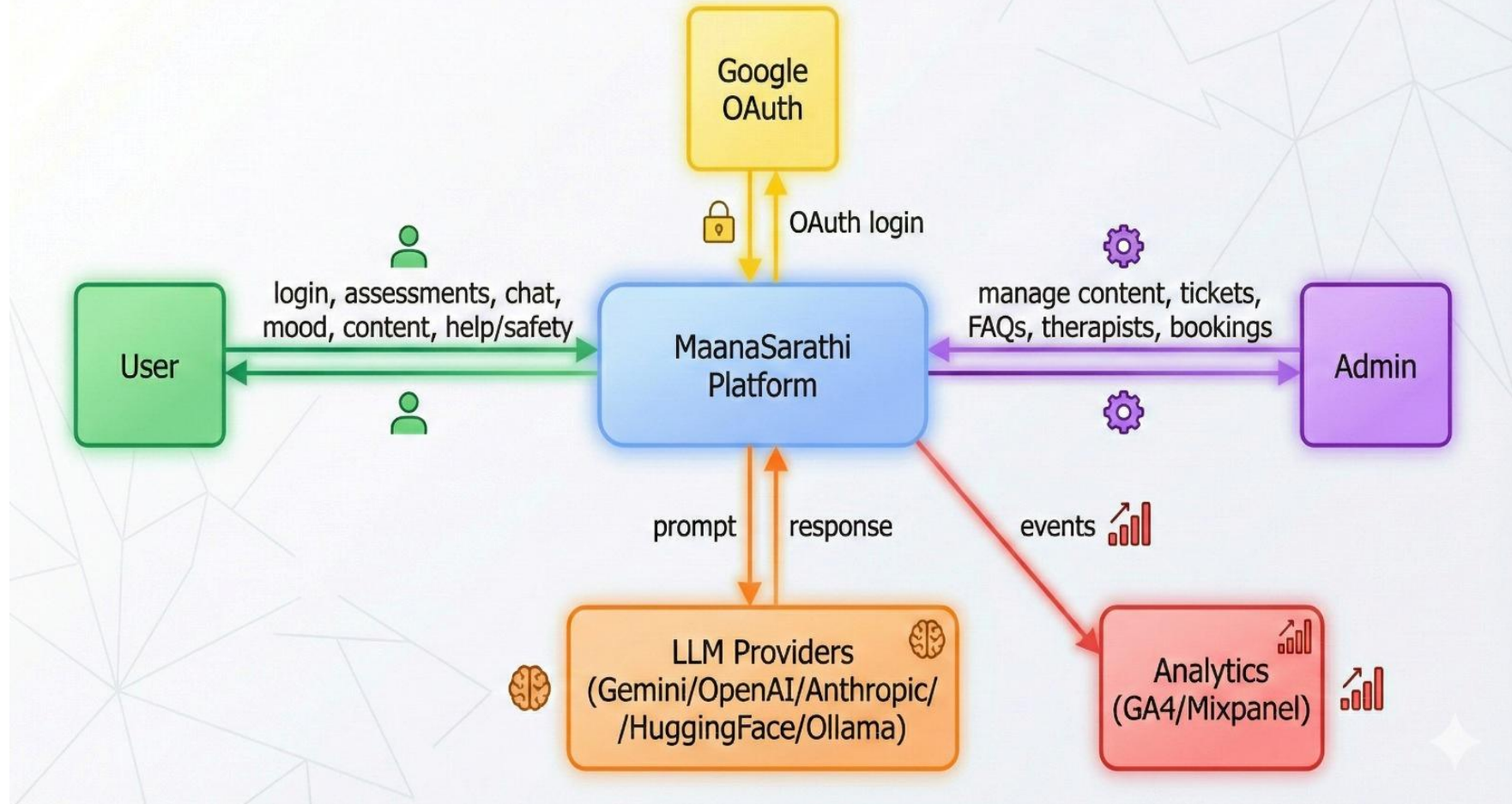
Proposed System

- **User Sign-Up:** Collect basic details (name, DOB, gender, contact info).
- **Initial Assessment:** Short mental health screening (under 5 minutes).
- **Optional Deep Tests:** Personality, anxiety, stress, etc.
- **AI Analysis:** Processes test results to determine wellbeing state.
- **Recommendation Path:** Western therapy, Eastern methods, or Hybrid.
- **Daily Support:** Chatbot, guided practices, curated content.
- **Progress Tracking:** Mood charts, engagement logs, reassessment prompts.

Data Flow Diagram

Level 0: System Context

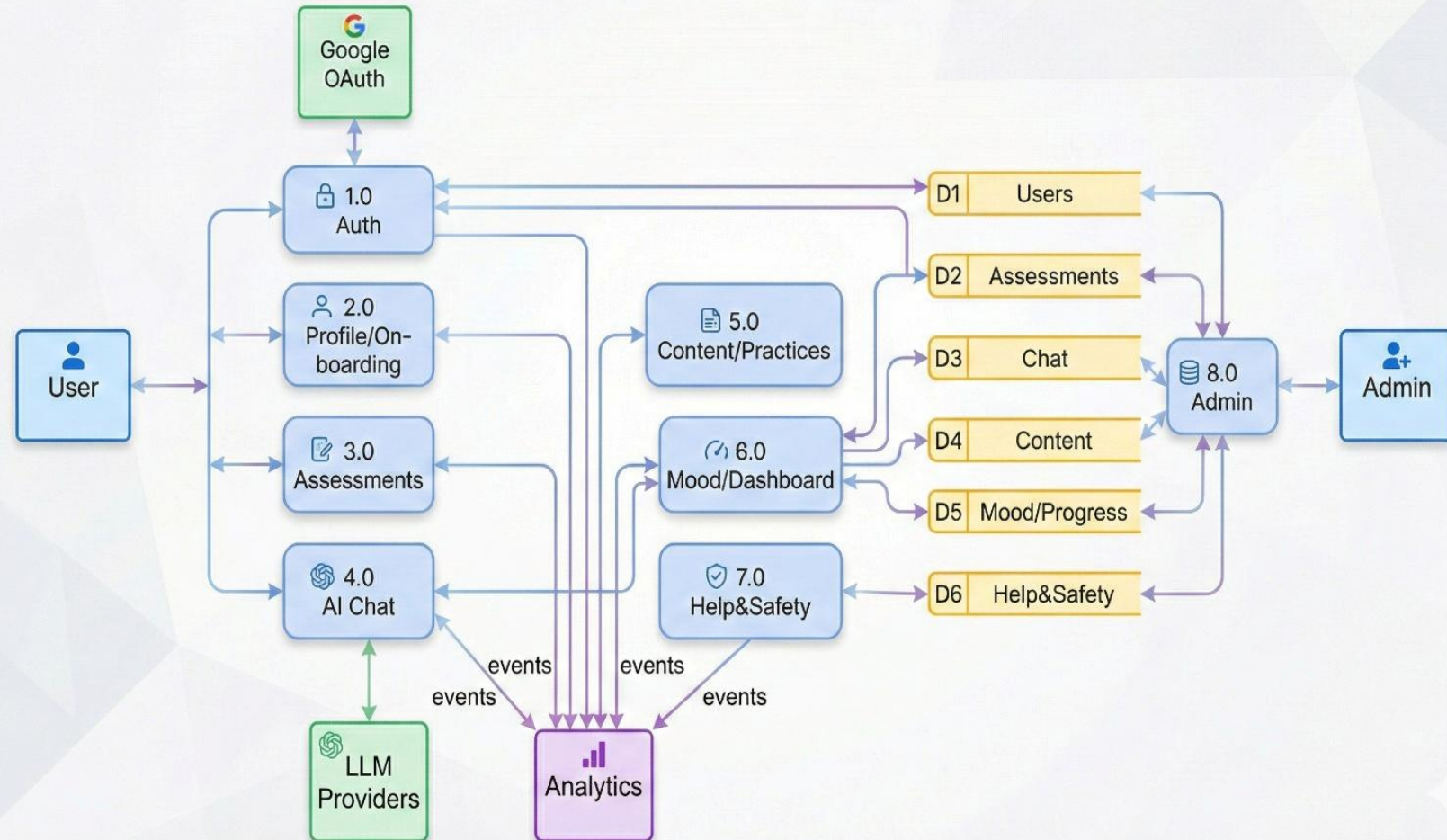
DFD Level 0 – MaanaSarathi



Data Flow Diagram

Level 1: Major Processes

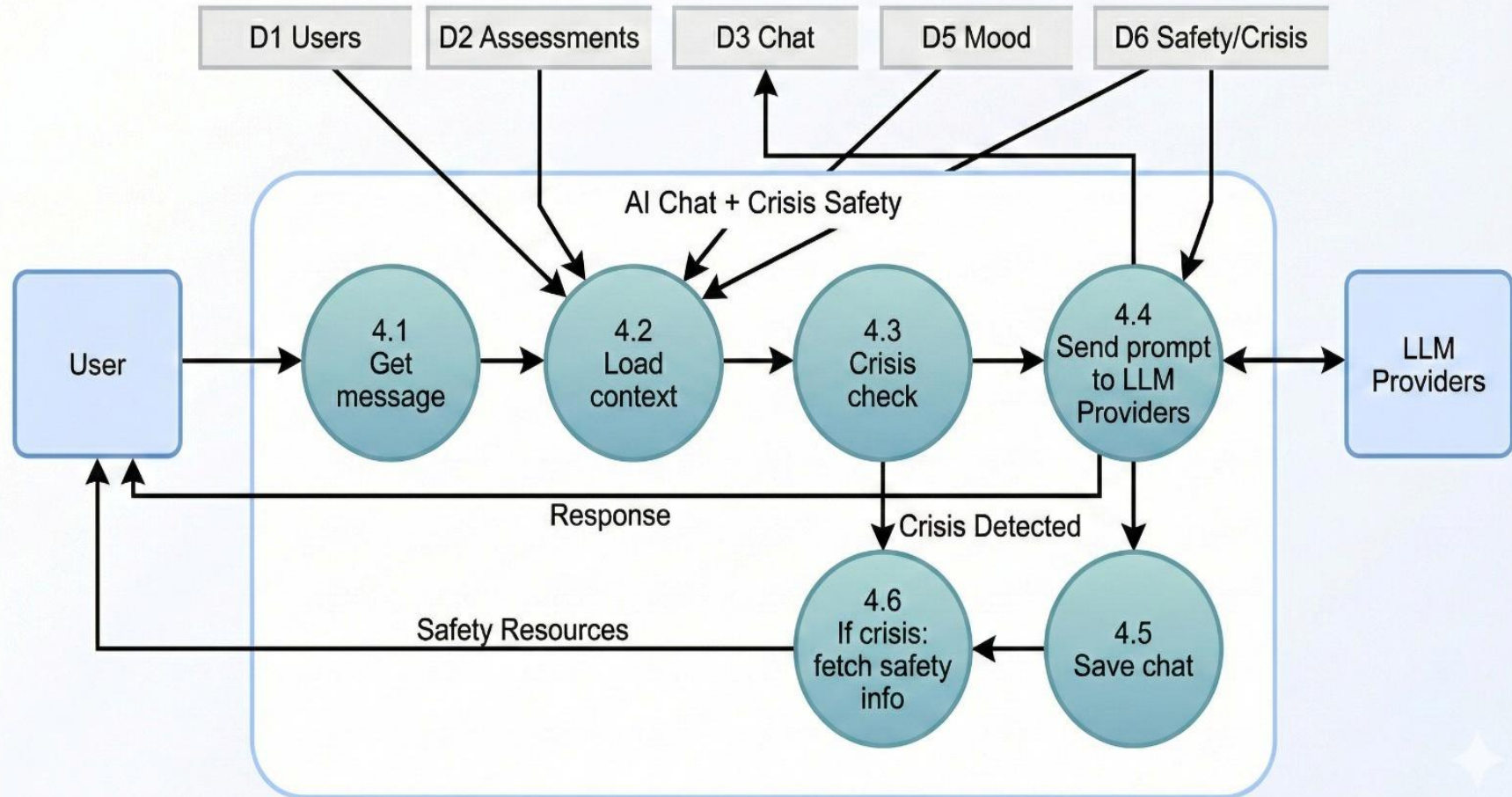
DFD Level 1 – MaanaSarathi



Data Flow Diagram

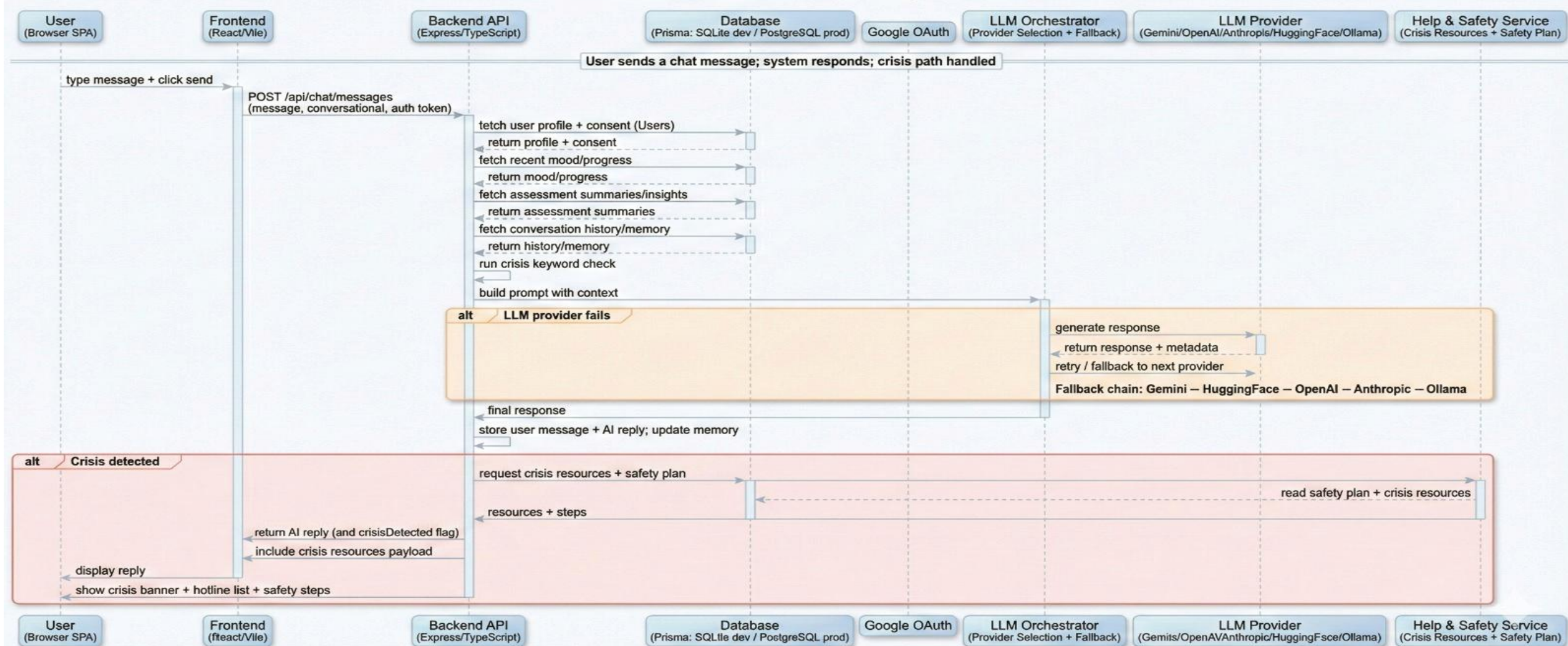
Level 2: AI Chat System

DFD Level 2 - AI Chat + Crisis Safety

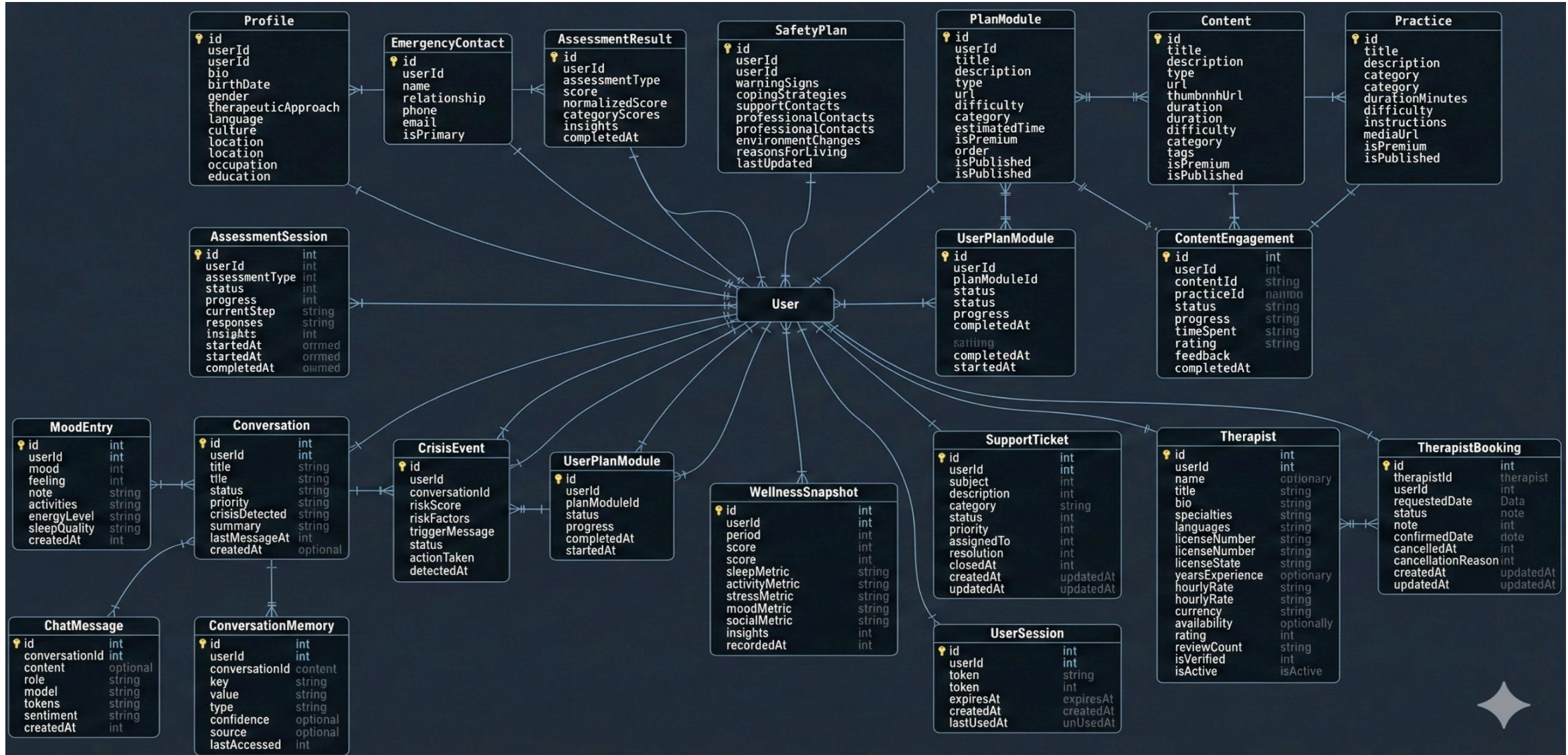


Sequence Diagram

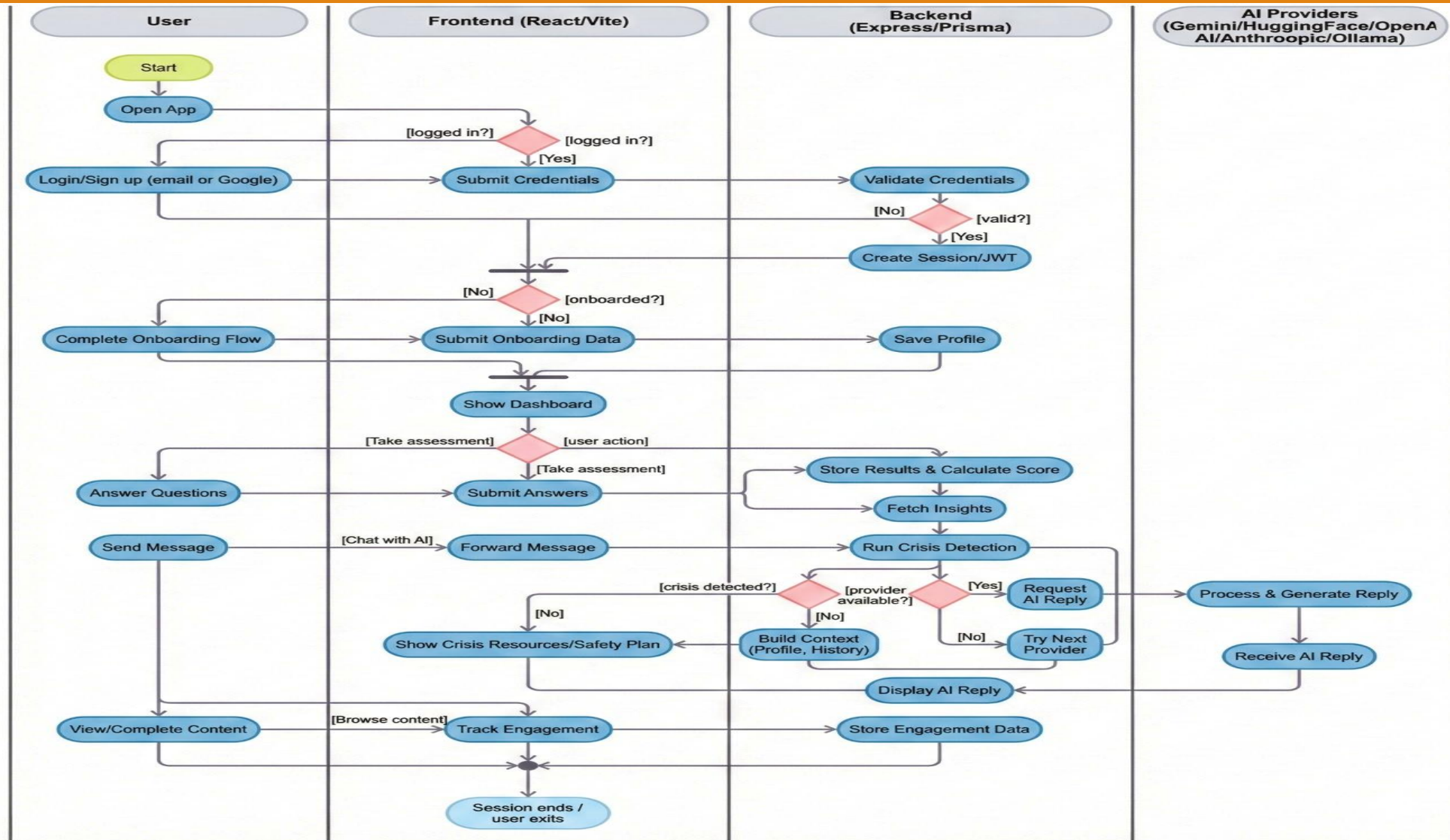
MaanaSarathi – AI Mental Wellbeing Platform



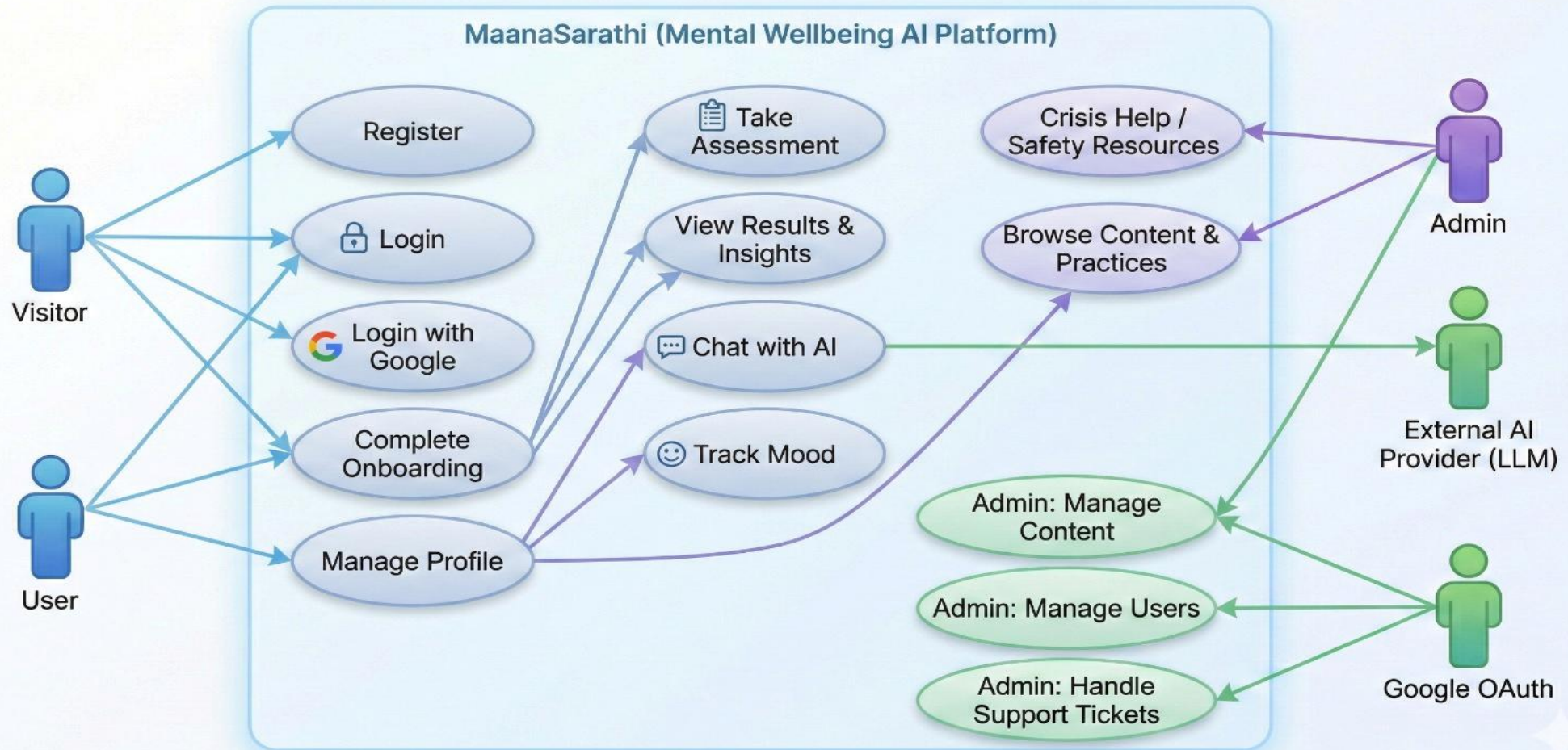
Class Diagram



Activity Diagram

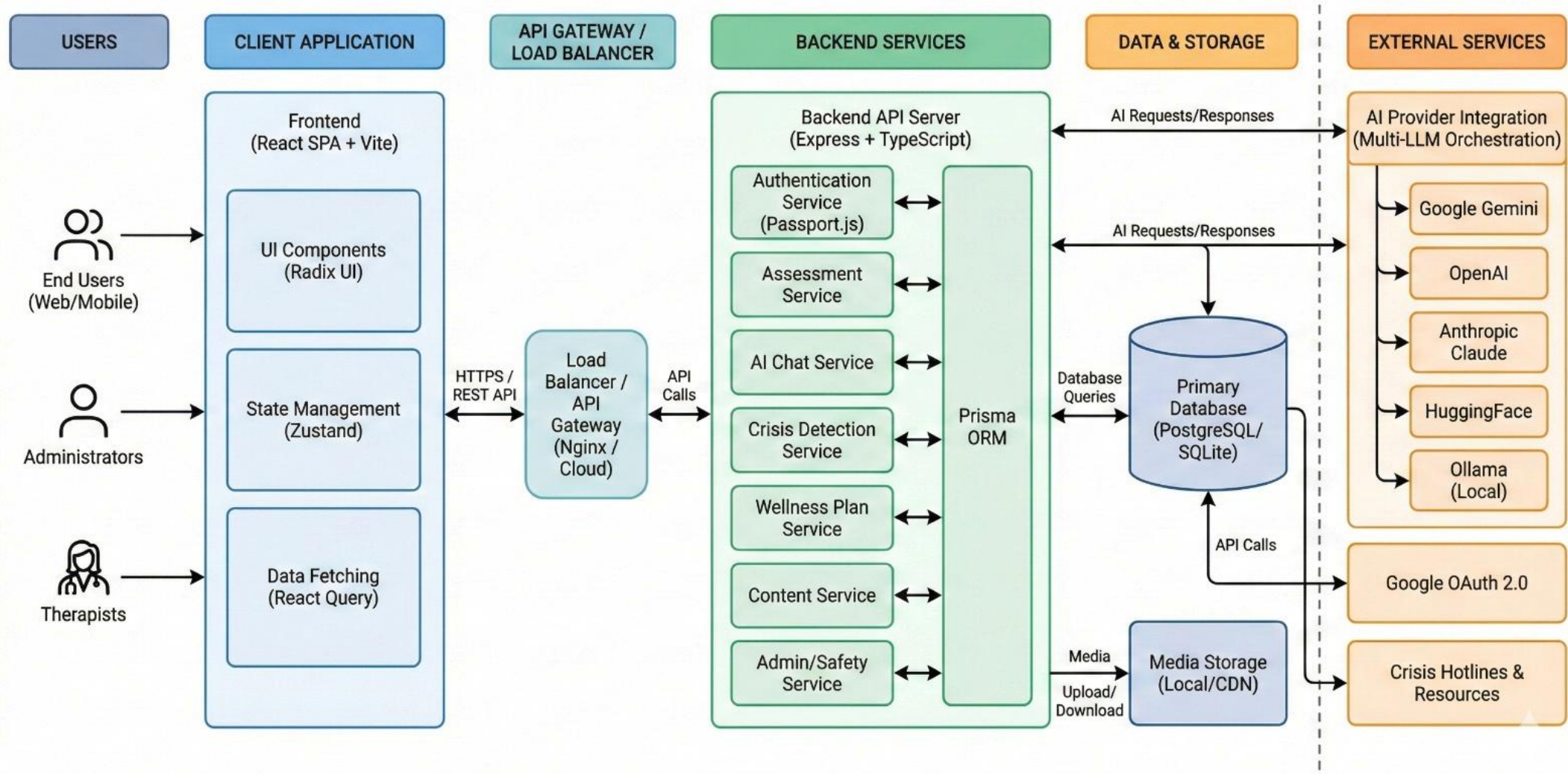


Unified Modeling Language.



Architecture Diagram

High-level System Architecture Dig for MaanaSarathi Mental Wellbeing AI App



OUTPUT-SCREENSHOT

← Back to Assessments

Session Standalone

PHQ-9 (Depression Assessment)

Patient Health Questionnaire-9 full depression inventory.

Question 2 of 9

~4 min remaining

22% complete

Feeling down, depressed, or hopeless

Not at all

Several days

More than half the days

Nearly every day

← Previous

Next →

Answered: 1 / 9

Time elapsed: 0 min

← Back

Local Admin

admin@example.com

Personal

Privacy

Notifications

Accessibility

Account

Personal Information

Edit Profile

First Name

Local

Last Name

Admin

Email Address

admin@example.com

Birthday

18 - 03 - 1985

Country / Region

Select country

Emergency Contact Name

Alex Taylor

Emergency Contact Phone

+1-555-201-3478

We'll only suggest contacting this person if our AI detects you might be in crisis. You always have full control.

Language Preferences

App Language

Choose your preferred language for the application interface

Language changes apply immediately and are saved automatically

Logout

← Back

If you are in immediate danger or having thoughts of self-harm:
Call 988 (US) • Text HOME to 741741 • Call 911 for emergencies

Call 988

Help & Safety

Resources, support, and answers to help you on your wellbeing journey

Crisis Resources

FAQ

Find Therapist

Contact Support

Find Therapist

My Bookings

Your Consultations (2)

Refresh

Dr. Sarah Johnson

Wed, Apr 1, 2026 11:30 AM

CONFIRMED

Phone Call

Cancel

Dr. Sarah Johnson

Sun, May 3, 2026 11:30 AM

CONFIRMED

Therapist Note
I will help you soon...

Video Call

Cancel

If you are in immediate danger or having thoughts of self-harm:
Call 988 (US) • Text HOME to 741741 • Call 911 for emergencies

Call 988

← Back

Help & Safety

Resources, support, and answers to help you on your wellbeing journey

Crisis Resources

FAQ

Find Therapist

Contact Support

Find Therapist

My Bookings

Find a Licensed Therapist

Connect with licensed professionals in your area. This directory includes verified therapists who specialize in various areas.

Specialty:

All

Anxiety

Depression

Stress Management

CBT

Mindfulness

Trauma

PTSD

EMDR

Grief

Relationship Issues

Medication Management

Bipolar Disorder

Psychiatric Evaluation

Couples Therapy

Family Therapy

Communication

Perinatal Counseling

OCD

Panic Disorder

Phobias

ERP

Anxiety Disorders

Dr. Sarah Johnson

Verified

Clinical Psychologist, PhD • PSYCHOLOGIST

5.0

San Francisco, CA

Dr. Johnson specializes in cognitive-behavioral therapy (CBT) and mindfulness-based interventions. With over 15 years of experience, she helps clients manage anxiety, depression, and stress. She creates a warm, non-judgmental space where clients can explore their thoughts and feelings.

Anxiety

Depression

Stress Management

CBT

Mindfulness

Call

Email

Website

Insurance Accepted

Request Consultation

← Back

Mindful Practices

Guided meditations, breathing exercises, and gentle yoga practices for your wellbeing

Search practices...

All Practices

Meditation

Breathing

Yoga

More

38 practices

15 min

ACS

15 Minute Body Scan

Led by Silvia Clark, MPH

Sleep onset script combining body scan, breath pacing, and stimulus-control reminders.

with Guide

Quick Relief

On-the-go

variety-apex

Sleep hygiene

Insomnia

Start Practice

7 min

BODY SHAKING

5 MIN FULL BODY RELEASE

Led by Silvia Clark, MPH

Somatic discharge movement for stress arousal release. Source: https://youtube.com/watch?v=y5JT7ESMDH4

with Guide

Quick Relief

variety-apex

movement

somatic

Start Practice

10 min

Variety Apex Practice: Identity and Boundary Reflection

Deep self-reflection practice for boundary clarity in emotionally demanding roles.

with Guide

variety-apex

self-reflection

boundaries

Start Practice

Good morning, Local Admin!

How are you feeling today?

Customize Dashboard

Profile

Logout

Quick mood check

Great

Good

Okay

Struggling

Anxious

Choose a mood to personalize your dashboard.

Quick Actions

Take Assessment
Get personalized insights

Chat with AI

Browse Library

View Progress

Recommended next step

Log today's mood

A quick check-in helps personalize your practices and insights.

Check in now

Today's Practice

Compassion pause for overall wellbeing

Start Practice

SOFTWARE / HARDWARE

Software:

- Frontend: React 18 & TypeScript
- Backend: Node.js + Express
- AI: OpenAI API, Gemini API, Hugging Face API, Local machine LLM
- Database: Firebase / MongoDB / Prisma DB / SQL lite
- Hosting: AWS / Google Cloud / local host
- Content Storage: Cloud storage with API integration (YouTube API)

Hardware:

- Mobile device (Android/iOS)
- Server infrastructure (cloud-hosted)
- Optional wearable integration for biometrics

Test Identification and Planning

Test Cases for Authentication Module:

- Test successful user registration with valid data
- Test registration with duplicate email (should fail)
- Test login with correct credentials
- Test login with incorrect password
- Test JWT token generation and validation
- Test session timeout handling

Test

Performance Testing:

- Load testing with 1000 concurrent users
- Stress testing chatbot response times
- Database query optimization testing

Security Testing:

- Penetration testing for API endpoints
- Encryption verification for sensitive data
- Authentication bypass attempts

PROJECT PLAN

Phase 1: Requirement gathering & research (2 weeks)

Phase 2: UI/UX design (3 weeks)

Phase 3: Core backend + AI integration (6 weeks)

Phase 4: Feature development (chatbot, tests, recommendations) (8 weeks)

Phase 5: Testing & feedback (4 weeks)

Phase 6: Deployment & monitoring (2 weeks)

Total: ~25 weeks (6 months) for completion.

SCOPE

Core Functionalities:

- WHO (World Health Organization) reports 1 in 8 people live with mental disorders; early detection improves outcomes.
- AI chatbots (e.g., Woebot, Wysa) have shown positive impact in reducing symptoms of anxiety and depression.
- Ancient practices like Yoga and Meditation have proven benefits for stress reduction and emotional stability.
- Hybrid approaches combining digital CBT (Cognitive Behavioral Therapy) and mindfulness show improved engagement and results.

Conclusion

Our Manasarathi AI Mental Wellness platform combines **scientific therapy** and **ancient wisdom** in a simple, accessible, and personalized platform. By delivering both emotional support and actionable guidance, it empowers users to take charge of their mental health anytime, anywhere.

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